



Mid-Mount Ladder Truck Workgroup

System Assessment and Validation for Emergency Responders: Workgroup Evaluation of Advanced Extrication and Breaching Technologies

APPROVED FOR INTERNAL DEPARTMENT RELEASE

Report Designation: SAVER-MBFD-EVAL-26

Publication Date: March 2026

Prepared For: Health and Safety Committee, Miami Beach Fire Department

Reviewing Authority: Chief Digna Abello





1.0 Front Matter and Administrative Declarations

1.1 Legal Notice and Disclaimer

The information, data, and findings contained within this document are intended exclusively for the internal strategic planning, procurement consideration, and operational assessment of the Miami Beach Fire Department (MBFD). The evaluation relies upon manufacturer-provided specifications, third-party laboratory certifications, and controlled workgroup testing. The inclusion of specific commercial off-the-shelf (COTS) products, trade names, trademarks, or manufacturer designations does not constitute an official endorsement, recommendation, or warranty of merchantability by the evaluating body.¹ The evaluating entity assumes no legal liability for the accuracy, completeness, or operational efficacy of the manufacturer-claimed specifications detailed herein, particularly those metrics which have not been subjected to independent National Fire Protection Association (NFPA) validation.¹

1.2 Foreword

The modern fire service operates within an increasingly complex and hazardous environment defined by rapid advancements in vehicle metallurgy, intricate building construction techniques, and the recognized physiological limitations of emergency responders. To maintain the elite operational standards required of an Insurance Services Office (ISO) Class 1 department and a Commission on Fire Accreditation International (CFAI) accredited agency, the continuous evaluation of rescue, extrication, and stabilization technology is an absolute necessity.²

The primary objective of this assessment is to answer two fundamental questions regarding the current commercial market: "What advanced rescue equipment is currently available?" and "How does this equipment perform relative to departmental health, safety, and tactical mandates?".¹ By systematically analyzing the technological landscape, this document serves to guide informed procurement decisions that protect the physical well-being of the responder while maximizing the survivability of the citizen.

1.3 Executive Summary

February of 2026, a specialized workgroup was convened to evaluate a cross-section of advanced rescue tools tailored for vehicle extrication, structural breaching, and incident stabilization. This market survey and evaluation report analyzes five specific tactical tools currently available in the commercial market, chosen for their potential to address acute operational challenges within the municipal response area.¹

The evaluation meticulously assessed technical specifications, ergonomic impacts, biomechanical strain reduction, and NFPA 1936 compliance.¹ The tools analyzed represent a diverse array of battery-operated, hydraulic, and manual breaching technologies, specifically including:

1. **Holmatro T1 Forcible Entry Tool:** A self-contained, multi-functional manual hydraulic



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- breaching device.³
2. **Hurst Jaws of Life SP 555E2 Spreader:** A high-capacity, battery-powered eDRAULIC extrication spreader.⁴
 3. **Holmatro CU 5050 i Inclined Cutter:** An ergonomically optimized, high-performance hydraulic cutter featuring New Car Technology (NCT) blades.⁵
 4. **DeWalt 60V MAX 12-Inch Cut-Off Saw:** A brushless, battery-powered concrete and steel breaching saw.⁶
 5. **Holmatro V-Strut:** A rapid-deployment vehicle stabilization system.⁸

The subsequent analysis reveals a definitive industry paradigm shift toward untethered, battery-powered operations, multi-functional capabilities, and ergonomically engineered geometries designed specifically to reduce the acute biomechanical load placed on human operators.⁹ By adopting equipment that integrates these technological advancements, emergency response agencies can achieve significantly faster patient access times while simultaneously mitigating the long-term musculoskeletal and respiratory injury risks inherent in heavy rescue operations.

2.0 Operational Context and Technological Overview

2.1 The Municipal Operating Environment

The jurisdiction protected by the evaluating agency presents unique geographic, demographic, and infrastructural challenges. The presence of coastal saltwater environments accelerates the corrosion and degradation of exposed metallic rescue components, necessitating tools with robust Ingress Protection (IP) ratings and anti-corrosive alloys.¹ Furthermore, the dense urban landscape, characterized by high-rise structures, subterranean parking garages, and congested causeways, severely limits the functional deployment of traditional tethered hydraulic systems.

Historically, hydraulic rescue operations required a direct physical connection to a heavy, combustion-engine power unit. This legacy framework presented multiple critical hazards. First, gas-powered hydraulic pumps routinely generate acoustic output exceeding 100 decibels (dBA), which inherently hinders critical fireground communication and contributes to cumulative sensorineural hearing loss among operators.¹¹ Second, the necessity of navigating high-pressure hydraulic hoses through dynamic, sharp-edged debris fields creates acute trip hazards and risks of hydraulic fluid injection injuries. Finally, the sequential deployment of pumps, hoses, and individual tool connections introduces an unavoidable latency period, delaying critical patient access during the "golden hour" of trauma survivability.

2.2 Shift to Untethered and Advanced Kinetic Systems

To resolve the limitations of tethered systems, the commercial rescue market has decisively shifted toward self-contained battery-powered tools (such as the Hurst eDRAULIC series) and advanced mechanical leverage systems (such as the Holmatro T1).¹² These innovations



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completely eliminate the hose tether, drastically reduce ambient noise profiles, and allow for immediate, single-operator deployment directly from the apparatus compartment.¹¹

The integration of high-density lithium-ion (Li-Ion) battery technology has enabled electric brushless direct current (DC) motors to generate the torque necessary to drive internal hydraulic pumps at performance levels indistinguishable from traditional combustion engines.¹

2.3 Modern Automotive Metallurgy and Extrication Challenges

The necessity for increasingly powerful extrication tools is driven by the rapid evolution of automotive manufacturing. In pursuit of enhanced passenger safety and fuel efficiency, the global automotive industry has aggressively integrated Ultra-High-Strength Steels (UHSS), Advanced High-Strength Steels (AHSS), and Boron steel alloys into the structural framework of modern vehicles.⁵

Specifically, the B-pillars, rocker panels, and roof rails of contemporary electric vehicles (EVs) and luxury sport utility vehicles (SUVs) present a formidable barrier to legacy cutting equipment. Boron steel, which is hot-stamped and rapidly cooled during manufacturing, boasts a tensile strength that can effortlessly fracture or shatter older, straight-blade rescue cutters.⁵ Consequently, modern rescue tools must utilize specialized blade geometries, such as U-shaped New Car Technology (NCT) blades, which are engineered to surround wide pillars and dynamically pull the dense material toward the tool's central pivot bolt, maximizing mechanical advantage and preventing the material from violently twisting out of the jaw.⁵

2.4 Ergonomics and Firefighter Health Implications

The evaluation of rescue tools must transcend raw kinetic power and rigorously analyze the biomechanical impact on the human user. First responders operating in structural or vehicular rescue environments are already burdened by the significant weight of their Personal Protective Equipment (PPE) and Self-Contained Breathing Apparatus (SCBA).¹⁵ Studies indicate that the added mass of PPE and SCBA fundamentally alters a responder's center of gravity, degrades functional balance, and increases cardiovascular strain during intermittent activities.¹⁵ Adding a 40-to-50-pound extrication tool to this existing physiological burden exacerbates musculoskeletal strain. Operating heavy, unbalanced tools outside the body's ergonomic "power zone"—specifically, holding equipment above the shoulders during a high roof-post cut, or crouching severely with flexed wrists to sever a rocker panel—exponentially increases shear forces on the lumbar spine and accelerates rapid muscle fatigue.¹⁰ Tools that optimize working angles, such as cutters with inclined jaw designs or spreaders with centralized mass distribution, directly mitigate this risk by allowing the operator to maintain a neutral spinal posture during complex extrications.¹⁰

2.5 Understanding NFPA 1936 Standards and Metrics

To protect emergency responders and ensure the reliability of municipal investments, theoretical performance calculations must be discarded in favor of standardized, empirical testing data. The National Fire Protection Association (NFPA) 1936, *Standard on Rescue Tools*, establishes the definitive testing parameters and design criteria for vehicle extrication



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equipment.¹

The standard exists to rectify historical inconsistencies where manufacturers advertised the "theoretical maximum force" of a tool. Theoretical force is calculated simply by multiplying internal hydraulic pressure by the surface area of the internal piston.¹⁸ However, because mechanical advantage fluctuates drastically as blades or arms articulate through their arc of motion, theoretical force does not accurately represent the tool's actual performance capabilities.¹⁸ As an example, the maximum theoretical force of a cutter is often achieved at the precise moment the blades cross and fully close—a point at which the force is entirely unusable for cutting a vehicle pillar.¹⁸

To provide a realistic comparison, NFPA 1936 mandates the measurement of specific, usable forces:

- **Highest Spreading Force (HSF) & Lowest Spreading Force (LSF):** Force is measured at ten equidistant points from the closed to the fully open position. The LSF is the most critical metric for procurement, as it represents the minimum guaranteed force the tool will apply at its weakest mechanical angle.¹⁸
- **Highest Pulling Force (HPF) & Lowest Pulling Force (LPF):** Measured similarly during the retraction of the spreader arms under load.¹⁸

Alphanumeric Cutter Performance Ratings: Cutter performance is validated by successfully severing specific, standardized high-strength metallic materials. The rating system utilizes a letter (A through F) to denote the material profile and a number (1 through 9) to denote the thickness or diameter of the material that the tool successfully severed.¹

- **A:** Round Bar (A-36 Hot-Rolled Steel)
- **B:** Flat Bar (A-36 Hot-Rolled Steel)
- **C:** Round Pipe (Schedule 40 Steel)
- **D:** Square Tube (A-36 Hot-Rolled Steel)
- **E:** Angle Iron (A-36 Hot-Rolled Steel)
- **F:** Rectangular Tube (UHSS / Boron equivalence)

A tool boasting an A8/B8/C7/D9/E9/F4 rating has been independently verified to cut extremely dense materials across all profiles, ensuring its capability against modern vehicle construction.¹

3.0 Workgroup Equipment Evaluations and Technical Specifications

This section provides an exhaustive technical and tactical analysis of the five specific tools identified for evaluation by the workgroup. The data presented is derived from manufacturer specifications and applicable third-party NFPA testing metrics.¹

!(Image 1)

Figure 3-1: Documentation from the preliminary workgroup analysis detailing performance requirements and market survey initial findings.

3.1 Holmatro T1 Forcible Entry Tool



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The Holmatro T1 represents a fundamental evolution in tactical breaching and forcible entry methodology. Rather than relying on a traditional, hydraulic “Rabbit tool”—the T1 consolidates six distinct tactical functions into a single, self-contained, manually operated hydraulic device.¹²



Figure 3-2: The Holmatro T1 Forcible Entry Tool, featuring an integrated hydraulic pump, spreader wedge, and cutting blades. Image Credit: Holmatro.

3.1.1 Technical Specifications

Specification Metric	Holmatro T1 Data
Tool Type	Manual Hydraulic Multi-Tool
Article Number	151.001.787 ³
Weight (Ready for Use)	17.0 lbs (7.7 kg) ³
Dimensions (L x W x H)	27.3 x 9.1 x 2.4 in (694 x 232 x 62 mm) ³
Maximum Working Pressure	10,443 psi (720 bar) ³
Maximum Spreading Distance	5.0 in (128 mm) ³
Maximum Spreading Force	7,419 lbf (33 kN) ³
Minimum Spreading Force (25mm)	5,845 lbf (26 kN) ³
Maximum Cutting Opening	1.1 in (29 mm) ³
Theoretical Cutting Force	31,248 lbf (139 kN) ³
Cutting Capacity (Round Bar)	0.7 in (18 mm) ³
Hydraulic Safety Factor	2:1 ²¹

3.1.2 Operational Analysis and Tactical Deployment

The T1 is engineered to execute six distinct operations: cutting, spreading, lifting, wedging, ramming, and hammering.¹² This multi-functional capacity allows a single operator to perform complex breaches that traditionally required a two-person team.

The core innovation of the T1 is its patented two-stage manual hydraulic pump. The application of just 66 pounds (30 kg) of manual force on the pump rod generates an immense 14.2 tons (31,248 lbf) of hydraulic cutting force and 3.4 tons (7,419 lbf) of hydraulic spreading force.¹² This massive mechanical transmission implies that operators expend significantly less cardiovascular energy and consume less breathing air from their SCBA compared to repeatedly striking traditional manual tools with an 8-pound sledgehammer.¹²

Tactically, the integration of a detachable wedge eliminates the critical need for a secondary operator to capture progress (holding the gap) during an inward or outward swinging door



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breach.¹² Weighing exactly 17 pounds, the T1 is sufficiently heavy to function effectively as a kinetic battering ram for light, inward-opening residential doors, yet balanced enough for the precision cutting of padlocks, chains, and up to 0.7-inch rebar.³ In complex search and rescue scenarios, where maintaining a free hand is critical for tactile victim location in zero-visibility environments, deploying with a single-operator breaching tool prevents the logistical bottleneck and physical exhaustion associated with carrying married irons.²⁰

3.2 Hurst Jaws of Life SP 555E2 Spreader

The Hurst SP 555E2 is an advanced, commercial-off-the-shelf, battery-powered extrication spreader utilizing proprietary eDRAULIC technology. Engineered with a single integrated cylinder body design, the SP 555E2 focuses on delivering massive structural displacement capabilities without the restrictions of a hydraulic hose tether.⁴



3.2.1 Technical Specifications

Specification Metric	Hurst SP 555E2 Data
Power Source	Proprietary eDRAULIC Li-Ion Battery (or 110V adapter) ⁴
Weight (Without Battery)	43.9 lbs (19.9 kg) ²³
Weight (With Battery)	44.1 lbs (20.0 kg) ⁴
Dimensions (L x W x H)	39.4 x 10.4 x 11.0 in (1,002 x 265 x 280 mm) ⁴
Spreading Distance	28.7 in (730 mm) ⁴
Max Spreading Force (Theoretical)	147,924 lbf (658 kN) ⁴
Max Pulling Force	13,039 lbf (58 kN) ⁴
NFPA Highest Spreading Force (HSF)	16,186 lbf (72 kN) ⁴
NFPA Lowest Spreading Force (LSF)	11,016 lbf (49 kN) ⁴
NFPA Highest Pulling Force (HPF)	10,341 lbf (46 kN) ⁴
NFPA Lowest Pulling Force (LPF)	6,925 lbf (28 kN) ⁴
Ingress Protection (IP) Rating	IP54 ⁴ / IP58 capability on EWXT variants ²³
Standard Compliance	NFPA 1936 / NFPA 1960 (2024) Compliant ⁴



3.2.2 Operational Analysis and Tactical Deployment

The SP 555E2 is highly regarded within the extrication community for achieving a 16% weight reduction while simultaneously delivering a 14% power increase over its direct predecessor (the SP 310E2).⁴

The most distinct tactical advantage of this spreader is the inclusion of heavily engineered, removable "Shark Tooth" tips. These specialized tips feature four rows of serrated, shark-like teeth designed to aggressively bite into high-tensile vehicular materials and structural components.⁴ In practical extrication scenarios—such as conducting a dashboard roll or popping a severely deformed door hinge—a spreader's theoretical maximum force of 147,924 lbf is entirely rendered moot if the tool slips out of the purchase point.⁴ The Shark Tooth geometry ensures that the NFPA-verified Lowest Spreading Force (11,016 lbf) is efficiently and relentlessly transferred into the vehicle structure without kinetic energy being lost to slippage or structural tearing.⁴

Furthermore, the tool incorporates squeezing plates built directly into the inner arms, expanding its utility beyond spreading to include the controlled crushing of pipes, structural members, or vehicle framing.⁴ The system operates on specific eDRAULIC batteries, which generally provide up to 50 minutes of continuous runtime and can be swapped rapidly on scene.¹ For protracted incidents, the unit can be connected to a 110V adapter for unstoppable, continuous power generation.⁴

3.3 Holmatro CU 5050 i Inclined Cutter

The Holmatro CU 5050 i is an elite hydraulic cutter engineered specifically to address the dual complexities of modern vehicle metallurgy and rescuer ergonomics. It utilizes a traditional external hydraulic power source connected via CORE technology—a highly efficient single-hose system containing both the high-pressure supply and low-pressure return lines within a coaxial configuration.⁵



Figure 3-4: The Holmatro CU 5050 i Inclined Cutter, highlighting the 30-degree offset jaw and integrated carrying handle with LED lighting. Image Credit: Holmatro.



3.3.1 Technical Specifications

Specification Metric	Holmatro CU 5050 i Data
Power Source	External Hydraulic Pump via CORE Technology ²⁵
Article Number	150.012.306 ⁵
Weight (Ready for Use)	37.7 lbs (17.1 kg) ⁵
Weight (Without Hose)	35.1 lbs (15.9 kg) ⁵
Dimensions (L x W x H)	30.0 x 12.0 x 10.6 in (762 x 306 x 268 mm) ⁵
Maximum Working Pressure	10,443 psi (720 bar) ⁵
Maximum Cutting Opening	7.2 in (182 mm) ⁵
Theoretical Cutting Force	312,260 lbf (1,389 kN) ⁵
Cutting Capacity (Round Bar)	1.6 in (41 mm) ⁵
NFPA 1936 Cutting Rating	A8 / B8 / C7 / D9 / E9 / F4 ⁵
EN 13204 Classification	BC165K-15.9 ⁵

3.3.2 Operational Analysis and Tactical Deployment

The CU 5050 i is structurally defined by two primary engineering innovations: the 30-degree inclined jaw and the implementation of New Car Technology (NCT) blades.

Ergonomic Redesign (The 30-Degree Incline): Traditional linear cutters force the operator to lift the entire mass of the tool (frequently exceeding 35 pounds) above shoulder level to execute high cuts, such as severing a vehicle's A-pillar or B-pillar at the roofline.¹⁰ Conversely, cutting low on a rocker panel requires severe crouching and unnatural flexion of the wrists. The 30-degree offset of the CU 5050 i resolves this biomechanical hazard. The angle allows the tool body to remain at a comfortable, neutral waist height while the blades properly address high and low cuts at the required 90-degree perpendicular angle to the target material.¹⁰ This design preserves operator stamina and dramatically reduces sheer stress on the lower back. Additionally, the inclined angle increases the physical working space between the tool body and the vehicle, minimizing the risk of the heavy tool pivoting inward and causing secondary blunt-force trauma to the victim during a violent relief cut.⁵

NCT and i-Bolt Technology: The NCT blades utilize a specialized U-shaped design that passively traps and pulls the target material deep into the recess of the jaw, drawing it directly toward the central pivot point where cutting force is exponentially highest.¹⁴ Coupled with this is the i-Bolt, a flat central bolt construction that actively squeezes the blades together during the progression of the cut.⁵ This prevents blade separation—a common failure point when older cutters attempt to sever modern Boron steel.⁵ This synergistic design directly translates to the tool's elite, third-party verified NFPA rating of A8/B8/C7/D9/E9/F4.⁵ Additional responder-focused features include an integrated carrying handle with powerful LED lighting to eliminate shadows during night operations, and an optimized control handle for superior



grip.²⁵

3.4 DeWalt 60V MAX 12-Inch Cut-Off Saw (DCS692)

As urban search and rescue (USAR), structural firefighting, and disaster response increasingly require the rapid breaching of concrete, masonry, asphalt, and reinforced steel, the DeWalt DCS692 provides internal-combustion-equivalent performance in a highly portable, zero-emission, battery-powered platform.²⁷



Figure 3-5: The DeWalt POWERSHIFT 12-inch Cut-Off Saw , demonstrating its compact profile, 5-position guard, and dual-sided water feed mechanism. Image Credit: DeWalt.

3.4.1 Technical Specifications

Specification Metric	DeWalt POWERSHIFT 12-inch Cut-Off Saw
Power Source	60V MAX FlexVolt Lithium-Ion Battery ²⁸
Motor Type	High-Efficiency Brushless DC ⁶
No-Load Speed	6,600 RPM ⁶
Weight (Bare Tool)	(Approx. 18.24 lbs operating)
Blade Diameter / Arbor	12-inch (230 mm) / 7/8 in ⁶
Maximum Depth of Cut	4.3/4 in (82.5 mm) at 90 Degrees ⁶
Dust Management	Dual-Sided Water Feed (OSHA Table 1 Compliant) ²⁸
Operational Controls	5-position rotatable guard, 3-position front handle ²⁸
Drivetrain	Beltless All-Metal Gearbox ²⁸

3.4.2 Operational Analysis and Tactical Deployment

The transition from traditional two-stroke gasoline rescue saws (colloquially known as "quickie saws" or K-12s) to a 60V high-draw battery ecosystem yields immediate tactical and health-related benefits. The battery 12" cut off saw entirely eliminates carbon monoxide emissions and particulate exhaust.²⁷ This allows rescue companies to deploy the saw deep within collapsed structures, subterranean trenches, or confined spaces without introducing toxic atmospheric conditions or requiring extensive positive pressure ventilation (PPV).²⁷



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Operating at a sustained 6,600 RPM and equipped with the 12-inch rescue blade yields a 4 3/4-inch maximum cut depth.⁶ This depth is precisely engineered for single-pass scoring of thick concrete slabs, curb cutting, and seamlessly severing heavy #3 through #8 structural rebar.⁶

An integrated heavy-load warning light provides the operator with instantaneous visual feedback, preventing the user from bearing down too hard, which mitigates blade binding and motor stall during aggressive breaching.⁶ Furthermore, the integration of a dual-sided water feed delivery system ensures total compliance with stringent OSHA silica dust regulations (Table 1), heavily mitigating long-term respiratory hazards and preventing silicosis among firefighters during structural collapse operations.²⁸

The saw features a sealed, latched battery box to protect the sensitive Li-Ion terminals from water, dust, and environmental degradation common on fire scenes.²⁸ The beltless, all-metal drivetrain eliminates the common failure point of snapped or slipping drive belts, reducing scene maintenance to zero.²⁸ Strategically, utilizing the DeWalt FlexVolt ecosystem allows the department to standardize high-capacity batteries across a fleet of chainsaws, blowers, and reciprocating saws, reducing logistical overhead and charger clutter on the apparatus (if we choose to utilize the 60v battery adapter).³²

3.5 Holmatro V-Strut Vehicle Stabilization System

Before hydraulic extrication kinematics can be applied safely, the chaotic accident scene must be rendered geometrically static. The Holmatro V-Strut is a rapidly deployable, tension-based shoring and stabilization system designed to establish a rigid structural triangle against severely compromised vehicles.³⁴



Figure 3-6: The Holmatro V-Strut, illustrating the integrated tensioning reel, auto-lock mechanism, and serrated multi-purpose head. Image Credit: Holmatro.



3.5.1 Technical Specifications

Specification Metric	Holmatro V-Strut Data
Article Number	150.062.158 ⁸
Weight (Ready for Use)	17.4 lbs (7.9 kg) ³⁴
Maximum Axial Load	3,597 lbf (16 kN / 1.6 t) ⁸
Minimum Effective Length	42.5 in (1,080 mm) ⁸
Maximum Effective Length	70.9 in (1,800 mm) ⁸
Dimensions (L x W x H)	42.5 x 5.9 x 8.3 in (1080 x 149 x 210 mm) ⁸
Deployment Time	~15 Seconds ³⁵

3.5.2 Operational Analysis and Tactical Deployment

The most perilous phase of a motor vehicle collision (MVC) response is the immediate arrival, where a multi-ton vehicle resting precariously on its roof, side, or an incline possesses massive potential energy. Any sudden shift during patient assessment or initial door-popping can crush trapped occupants or severely injure responders.

The V-Strut resolves this dynamic threat by consolidating all necessary stabilization components—the extending strut body, a serrated multi-purpose engagement head, a wide non-slip base plate, and a high-strength tensioning belt featuring a hook and ratchet mechanism—into a single, unified 17.4-pound package.³⁵ By eliminating loose parts (such as separate chains, wedges, or pins that can be lost in the dark or mud), deployment latency is virtually eradicated.

The tool features a unique auto-lock mechanism that operates without a secondary locking pin. The firefighter simply pulls the strut to the required length, and it automatically engages locking holes set at granular intervals for highly fine-tuned positioning.³⁷ Total setup can routinely be achieved in under 15 seconds.³⁵

Once the serrated head aggressively engages the vehicle chassis or a purchase point, the operator unspools the integrated tensioning belt, attaches the snap hook to the vehicle's undercarriage, and ratchets the belt tight. This locks the system via tension triangulation, safely transferring the vehicle's load directly into the ground.³⁴ A specialized squeeze-and-push mechanism ensures the system can be easily depressurized and retracted following the conclusion of the incident.³⁷ Despite weighing only 17.4 pounds, the V-Strut can support a massive axial load of 16 kN, rendering it capable of stabilizing heavy passenger vehicles, SUVs, and light commercial trucks.³⁵



4.0 Workgroup Comparative Analysis and Findings

To synthesize the raw specifications into actionable intelligence for the MBFD Health and Safety Committee, the evaluated tools were cross-analyzed against three core departmental priorities: deployment speed, operator health and safety (ergonomic risk management), and infrastructural interoperability.

4.1 Advancements in Ergonomic Risk Mitigation

The evaluation unequivocally demonstrates a direct correlation between advanced tool geometry and the reduction of cumulative physiological trauma among firefighters. The fire service has historically accepted joint degradation and lumbar injuries as inherent risks of heavy rescue. The integration of the tools evaluated here actively engineers those risks out of the operation.

- The **Holmatro CU 5050 i** mathematically alters the fulcrum of lifting required by the operator. By integrating a 30-degree tilt on the cutter head, the firefighter avoids extreme spinal flexion and shoulder hyperextension, keeping the 37.7 lb mass centered close to the body's natural power zone.⁵ Over a 25-year career, the reduction in sheer stress on the L4/L5 vertebrae is incalculable.
- The **Holmatro T1** acts as a force multiplier for a single operator. By replacing a 10-15 lb married irons set, a sledgehammer, and a secondary hydraulic rabbit-tool pump, the rescuer remains highly mobile. This reduces cardiovascular expenditure and air consumption prior to entering a compromised, immediately dangerous to life or health (IDLH) environment.³

4.2 Standardization of Power Ecosystems and Logistical Footprint

A significant operational divide exists between proprietary battery ecosystems and commercial off-the-shelf (COTS) architectures.

- The **Hurst SP 555E2** utilizes proprietary eDRAULIC lithium-ion batteries. While these provide incredibly robust, specialized energy delivery with an IP54 environmental rating tailored for extreme temperatures, they lock the agency into a closed lifecycle matrix. The chargers and battery cells cannot be cross-purposed for other fireground tools.⁴ This necessitates dedicated compartment space and specific budget line items for proprietary replacements, which historical data suggests can exceed \$300 per unit.³²
- Conversely, integrating tools like the **DeWalt DCS692** utilizes a 60V MAX FlexVolt COTS ecosystem. This strategy allows the department to deploy a singular, interchangeable battery architecture that powers concrete saws, ventilation chainsaws, positive pressure fans, and scene lighting. This approach aggressively minimizes capital expenditure on backup batteries, drastically reduces spatial requirements and charger clutter on the apparatus, and ensures long-term availability of power units through commercial hardware channels.³²



4.3 Force Application vs. Theoretical Power

Data from the NFPA 1936 testing confirms that raw theoretical power is secondary to applied mechanics and blade geometry.

- The **Hurst SP 555E2** utilizes its heavily engineered "Shark Tooth" geometry to ensure its 16,186 lbs of NFPA-verified HSF is not lost to material slippage.⁴ The teeth act as localized stress concentrators, piercing the metal to lock the tool in place while spreading force is applied.
- The **Holmatro CU 5050 i** utilizes the i-Bolt and NCT U-shaped blades to prevent blade separation when addressing ultra-high-strength steel.⁵ Both tools prove conclusively that capturing, holding, and manipulating modern vehicle metallurgy requires specialized mechanical design, not merely increased internal hydraulic pressure.

5.0 Strategic Conclusions and Recommendations

Based on the exhaustive analysis of technical specifications, biomechanical impacts, NFPA standards, and operational workflows, the following strategic conclusions are presented to Chief Digna Abello and the MBFD Health and Safety Committee:

1. **Prioritize Ergonomic Geometries in Future Procurement:** The adoption of non-linear tool designs, specifically the 30-degree inclined jaw of the Holmatro CU 5050 i, should become a primary baseline criterion in all future extrication tool purchasing matrices. The measurable reduction in acute biomechanical stress directly aligns with the department's mandate to enhance firefighter health, reduce worker's compensation claims, and extend career longevity.¹⁰
2. **Transition Secondary Breaching Tools to COTS Battery Ecosystems:** Transitioning secondary tactical tools—such as rotary cutoff saws, ventilation chainsaws, and light breaching tools—to a unified, high-voltage COTS ecosystem (such as the DeWalt 60V FlexVolt system) offers unparalleled logistical efficiency. It resolves apparatus charger clutter, reduces lifetime battery replacement costs, and crucially mitigates the respiratory risks and acoustic trauma associated with two-stroke combustion engines operating in confined spaces.²⁷
3. **Strictly Mandate NFPA 1936 Verified Metrics:** When evaluating and comparing primary heavy extrication equipment, procurement documents must strictly utilize third-party verified NFPA metrics (HSF, LSF, and the A-F cutting capability ratings) rather than theoretical maximum forces.¹ This ensures objective performance comparisons and guarantees the tools possess the verified kinetic energy to defeat modern Boron steel.
4. **Integrate Self-Contained Multi-Tools for Rapid Primary Search:** Supplying initial attack and primary search teams with consolidated kinetic tools like the Holmatro T1 significantly enhances operational agility. By reducing the physical footprint and weight of traditional multi-piece breaching kits, a single operator can affect forcible entry more rapidly and with less exertion. This directly decreases the critical time-to-patient metric



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while preserving the operator's physical stamina for the subsequent search.¹²

5. **Adopt Rapid-Deployment Tension Stabilization Systems:** The integration of all-in-one stabilization mechanisms, such as the Holmatro V-Strut, entirely eliminates the latency and complexity of assembling disparate cribbing blocks and strapping components. The ability to achieve definitive structural stability within a 15-second operational envelope provides an immediate safety barrier, protecting both the trapped victims and the medical personnel interacting with the compromised vehicle.³⁵

The synthesis of untethered battery-powered mobility, advanced ergonomic geometric design, and zero-emission operation represents the modern standard for rescue technology. Aligning the Miami Beach Fire Department's procurement strategy with these technological innovations will directly support the agency's unyielding commitment to optimal public safety, fiscal responsibility, and premier operational accreditation.

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